

Raj Sudharshan Vidyalatha Natarajan

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(Open to Relocate)

Dynamic **Python Developer** with over **3+** years of **experience** in **Backend development** and **Cloud technologies**, specializing in crafting scalable cloud architectures and optimizing system performance. **AWS Certified Solutions Architect** with proven expertise in designing and implementing robust cloud solutions that leverage the full suite of AWS services. Proficient in programming languages and frameworks such as **Python**, **Django**, and **JavaScript**. **Actively seeking opportunities** to leverage expertise and contribute to organizational success in a challenging role

AWARDS, RECOGNITION, CERTIFICATIONS

- “Go That Extra Mile Award” by Wipro
- 2nd place, Google Developer Student Club Hackathon
- [AWS Certified Solutions Architect – Associate](#) (Amazon Web Services)
- [AWS Academy Graduate - AWS Academy Cloud Architecting](#)
- [Django Celery Mastery – Python Asynchronous Task Processing](#) (Udemy)

PROFESSIONAL EXPERIENCE

SIMPALM

Python Developer

Client: BUDY

Software Engineer

San jose,CA

Feb 2024 - Feb 2025

At Budy, a platform leveraging AI agents for seamless data exchange between top CRMs like Salesforce and HubSpot, I implemented a robust AAA (Authentication, Authorization, and Accounting) system to enable secure, role-based data transfers. Additionally, I built a comprehensive CI/CD pipeline to streamline deployments and accelerate feature rollouts

- Migrated authentication to **Google OAuth**, implementing custom **JWT** lifecycle management and payloads with **Redis** caching, reducing authentication processing time by **30%**
- Architected and deployed high-performance, distributed backend systems using **Python** achieving 40% faster API responses and supporting millions of concurrent requests
- Utilized **Django's** ORM to efficiently manage complex database interactions, ensuring high performance and reliability in data handling processes while maintaining data integrity
- Integrated **FastAPI** with asynchronous programming techniques, significantly improving response times and overall efficiency of data handling operations across the platform
- Developed and optimized **RESTful** and **GraphQL** APIs with Django REST Framework (DRF) and Strawberry, reducing API payload size by **35%** and improving front-end rendering speeds
- Built a **lambda**-tolerant, serverless data processing pipeline using **AWS Lambda**, Step Functions, and S3, reducing operational costs by 50% while processing terabytes of data
- Designed and implemented a **Kafka**-based microservices architecture for reliable inter-service communication, enabling robust error handling and high-throughput message processing
- Streamlined deployments via **AWS CodePipeline** with a blue-green deployment strategy, reducing downtime by **90%**
- Orchestrated container orchestration using **Docker** Swarm, optimizing resource allocation and improving fault tolerance by **40%**
- Migrated **Docker** volumes to **Amazon EFS** with custom IAM policies for secure and persistent data storage across multiple instances
- Improved scalability by integrating **load balancers** and **auto-scaling** groups with **EC2 instances**, enhancing system availability under high-traffic conditions

Client: VISIONIFY INC

Full Stack Developer

At Visionify, an AI/ML-based workplace safety solutions provider, I transformed an early-stage pilot prototype into a robust enterprise-grade system by developing real-time alerting and reporting features, integrating a monitoring dashboard, and implementing video recording and speaker integrations for enhanced safety notifications

- Designed and implemented scalable backend systems while ensuring system resilience and high availability across cloud and on-premise environments
- Developed backend architecture from scratch for three applications using **FastAPI**, **Redis**, **React**, and **MySQL**, delivering scalable and efficient systems
- Implemented **Celery** and Celery Beat with multiprocessing and threading for concurrency, optimizing task execution and improving throughput by **35%**
- Built observability solutions using Prometheus and Grafana, increasing system monitoring and response efficiency by 50%
- Enhanced a resilient data synchronization mechanism for edge and on-premises deployments, ensuring 99% data integrity during internet outages and reducing data loss incidents by 95%
- Orchestrated multi-environment deployments (edge server, on-premises, and cloud) using **Docker** and **Azure** Kubernetes Service (AKS), cutting deployment time by 40%
- Optimized system performance with Redis for caching and message queuing, reducing API response times by 60% and maintaining 99.9% availability during peak loads
- Developed and integrated frontend features using **React**, Redux, and React Query, reducing page load times by 45% and improving user experience
- Developed a real-time notification system for safety and compliance alerts using **Redis** and Celery, enabling automated voice announcements via integrated speaker systems and reducing manual monitoring

WIPRO TECHNOLOGIES

Software Developer

Client: Status Solutions USA

Chennai, India

May 2019 - Dec 2021

Led the migration of a retirement home community software system from PHP to Django, overseeing the transition as the lead developer. The software, which collected data from IOT devices patient wristbands, pendants, and other end devices, provided real-time monitoring and triggered alerts to nurses in case of any incidents

- Led the modernization of legacy systems, improving efficiency, scalability, and performance while reducing operational overhead
- Spearheaded migration from legacy **PHP** to **Django**, improving system performance by **60%** for over 10,000 daily users
- Developed an auto-assignment mechanism for patient-device matching, reducing manual intervention by 80% and cutting operational costs by 25%
- Implemented a real-time critical care tracking system using **Django** and **WebSockets**, improving response times by 40% and ensuring seamless integration with **IoT devices** (watches, pendants, sensors)
- Improved database query efficiency by 35% through the implementation of filter sets and advanced querying techniques within the **MVC framework**, enhancing data retrieval speed and system responsiveness under high-load conditions

Client: Mastercard

Software Developer

Managed and optimized financial transaction processing systems, ensuring high availability and compliance with industry regulations

- Managed the Automated Billing Unit (ABU) application, processing **100,000+** transactions daily and enhancing overall system efficiency by **30%**
- Key contributor in a global L3 support team, conducting in-depth code analysis and resolving **55+** high-priority issues to maintain a **99%** application uptime
- Implemented performance enhancements and security upgrades, ensuring compliance with financial industry standards

NEYVELI LIGNITE CORPORATION

Electrical Engineer Intern

Chennai, India

May 2016 - Aug 2016

At Neyveli Lignite Corporation, I worked closely with senior engineers and managers to gain a comprehensive understanding of large-scale electrical systems and their role in power generation and distribution

- Worked in Lignite Handling systems learning the workings of boilers, and electrical equipment including HVAC systems, transmission, transformers, etc
- Assisted in monitoring and optimizing power distribution processes to enhance system efficiency and reliability.

PROJECTS

Cloud Infrastructure Automation and Deployment

- Automated cross-cloud infrastructure deployment with Pulumi, deploying **AWS** services (EC2, RDS, VPCs) and integrating **GCP** Cloud Storage buckets for a unified multi-cloud strategy
- Created golden AMIs via Packer, integrated with GitHub Actions **CI/CD** pipeline for automated testing & deployment
- Developed **REST APIs** with integrated health checks, leveraging CloudWatch for in-depth logging and metrics analysis, and resource optimization
- Designed a scalable and high-performance environment with EC2 auto-scaling and Elastic Load Balancing, using **infrastructure-as-code (IAC)** methodologies
- Refined RDS configurations using security and parameter groups, leveraging user scripts for enhanced performance
- Deployed serverless architecture using Lambda, integrated with SNS for notification and **DynamoDB** for data processing with artifacts stored in **GCP** Cloud Storage for multi-cloud integration

Hashtag Holidays Chatbot

- Designed and developed an AI-powered chatbot to deliver personalized travel recommendations by analyzing Instagram activity and browsing data
- Leveraged Machine Learning (ML) and Natural Language Processing (NLP) to enhance the accuracy of recommendations, increasing user engagement
- Architected a responsive backend using **FastAPI** and **DialogFlow**, ensuring seamless interaction between the chatbot and users
- Deployed the chatbot on the Telegram platform, providing users with an intuitive, real-time travel assistance experience
- Implemented a dynamic scoring model using user behavior data and Google Maps, offering tailored travel suggestions based on real-time location insights

DineQ (Square Devpost Hackathon)

- Revolutionized dining with DineQ app (Django, Flutter, AWS, Postgres), to enhance the user experience and restaurant operations
- Slashed wait times by 30% by replacing physical queues with virtual ones, significantly boosting user satisfaction
- Implemented seamless, secure transactions by integrating real-time restaurant data, streamlining order processing, payments, and inventory management using Square API and Unique-Order-ID

Real-Time Cloud Data Pipeline

- Constructed a scalable, serverless cloud-native data pipeline on **AWS** to handle millions of real-time transaction data
- Set up a queue-triggered system for efficient data processing using AWS services like API Gateway, SQS, and Lambda
- Devised a system based on CloudWatch rules to transfer data from RDS to Redshift every 15 minutes, enabling real-time data availability for effective analysis and reporting

SKILLS

- **Databases & Caching:** PostgreSQL, MySQL, Redis, MongoDB, MSSQL
- **Programming Languages/OS:** Python, JavaScript, TypeScript, Java, SQL, NoSQL, Linux/Unix
- **Frameworks:** Django, Flask, FastAPI, React, Node.js, NestJS, Next.js
- **Tools & Libraries:** REST/RESTful APIs, Kafka, OAuth
- **Cloud & DevOps:** EC2, S3, CloudWatch, CloudFormation, RDS, Lambda, SQS, SNS, EFS, DynamoDB, Pulumi, Packer, Docker, Kubernetes, Terraform, GitHub Actions, CI/CD

EDUCATION

NORTHEASTERN UNIVERSITY

MS in Computer Software Engineering (GPA: 3.89/4)

California, US
Jan 2022 - Dec 2023

ANNA UNIVERSITY

BE in Electrical and Electronics Engineering

Chennai, India
2015 - 2019